

1.

MMU helps in:

- A. File transfer
- B. Address translation
- C. Scheduling tasks
- D. Disk formatting

Answer: B

2.

Internal fragmentation occurs when:

- A. Memory is too small for any process
- B. Memory block is larger than required
- C. Memory is divided in unequal parts
- D. Process size is larger than the memory block

Answer: B

3.

A computer has a memory access time of 100 nanoseconds. The time taken to handle a page fault is 5,000 nanoseconds, if the page fault rate is 0.01, what is the effective access time?

- A. 144 ns
- B. 140 ns
- C. 142 ns
- ✓ D. 149 ns

Answer: D

$$\begin{aligned}
 \text{EAT} &= (1 - \text{Page fault rate}) \times \text{memory access time} + \text{Page fault rate} \times \text{page fault service time} \\
 &= (1 - 0.01) \times 100 + 0.01 \times 5000 \\
 &= 0.99 \times 100 + 50 \\
 &= 99 + 50 = 149
 \end{aligned}$$

4.

Which file system is commonly used in Windows?

- A. EXT3**
- B. NTFS**
- C. HFS**
- D. XFS**

Answer: B

5.

Which of the following is an example of a disk storage device?

- A. RAM**
- B. CPU**
- C. Hard Disk Drive (HDD)**
- D. Monitor**

Answer: C

6.

Which allocation method is fastest in practice?

- A. Best Fit**
- B. Worst Fit**
- C. First Fit**
- D. Next Fit**

Answer: C

7.

In segmented memory management, which causes a segmentation fault?

- A. Address within segment limit**
- B. Address greater than limit**
- C. Address equal to base**
- D. Address in valid range**

Answer: B